CONCEPT 44.5

Hormonal circuits link kidney function, water balance, and blood pressure *(pp. 994–996)*

- The posterior pituitary gland releases **antidiuretic hormone (ADH)** when blood osmolarity rises above the normal range, such as when water intake is inadequate. ADH increases the permeability to water of the collecting ducts by increasing the number of epithelial **aquaporin** channels.
- When blood pressure or blood volume in the afferent arteriole drops, the **juxtaglomerular apparatus** releases renin. Angiotensin II formed in response to renin constricts arterioles and triggers release of the hormone aldosterone, raising blood pressure and reducing the release of renin. This **renin-angiotensin-aldosterone system** has functions that overlap with those of ADH and are opposed by **atrial natriuretic peptide**.

? *Why can only some patients with diabetes insipidus be treated effectively with ADH?*

TEST YOUR UNDERSTANDING

For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. *Unlike* an earthworm's metanephridia, a mammalian nephron
   (A) is intimately associated with a capillary network.
   (B) functions in both osmoregulation and excretion.
   (C) receives filtrate from blood instead of coelomic fluid.
   (D) has a transport epithelium.

2. Which process in the nephron is *least* selective?
   (A) filtration
   (B) reabsorption
   (C) active transport
   (D) secretion

3. Which of the following animals generally has the lowest volume of urine production?
   (A) vampire bat
   (B) salmon in fresh water
   (C) marine bony fish
   (D) freshwater flatworm

Levels 3-4: Applying/Analyzing

4. The high osmolarity of the renal medulla is maintained by which of the following?
   (A) active transport of salt from the upper region of the descending limb.
   (B) the loose packing of juxtamedullary nephrons.
   (C) diffusion of urea into the collecting duct.
   (D) diffusion of salt from the descending limb of the loop of Henle.

5. In which of the following species should natural selection favor the highest proportion of juxtamedullary nephrons?
   (A) a river otter
   (B) a mouse species living in a temperate broadleaf forest
   (C) a mouse species living in a desert
   (D) a beaver

6. African lungfish, which are often found in small, stagnant pools of fresh water, produce urea as a nitrogenous waste. What is an advantage of this adaptation?
   (A) Urea takes less energy to synthesize than ammonia.
   (B) Small, stagnant pools do not provide enough water to dilute ammonia, which is toxic.
   (C) Urea forms an insoluble precipitate.
   (D) Urea makes lungfish tissue hypoosmotic to the pool.

Levels 5-6: Evaluating/Creating

7. **INTERPRET THE DATA** (a) Use the data below to draw four pie charts for the average daily water gain and loss in a kangaroo rat and a human.

	Kangaroo Rat	Human
Average Water Gain (mL/day)		
Ingested in food	0.2	750
Ingested in liquid	0	1,500
Derived from metabolism	1.8	250
Average Water Loss (mL/day)		
Urine	0.45	1,500
Feces	0.09	100
Evaporation	1.46	900

   (b) Which routes of water gain and loss make up a much larger share of the total in a kangaroo rat than in a human?

8. **EVOLUTION CONNECTION** Merriam's kangaroo rats (*Dipodomys merriami*) live in North American habitats ranging from moist, cool woodlands to hot deserts. Based on the hypothesis that there are adaptive differences in water conservation between *D. merriami* populations, predict how the rates of evaporative water loss would differ for populations that live in moist versus dry environments. Propose a test of your prediction, using a humidity sensor to detect evaporative water loss by kangaroo rats.

9. **SCIENTIFIC INQUIRY** You are exploring kidney function in kangaroo rats. You measure urine volume and osmolarity, as well as the amount of chloride (Cl^-) and urea in the urine. If the water source provided to the animals were switched from tap water to a 2% NaCl solution, indicate what change in urine osmolarity you would expect. How would you determine if this change was more likely due to a change in the excretion of Cl^- or urea?

10. **WRITE ABOUT A THEME: ORGANIZATION** In a short essay (100–150 words), compare how membrane structures in the loop of Henle and collecting duct of the mammalian kidney enable water to be recovered from filtrate in the process of osmoregulation.

11. **SYNTHESIZE YOUR KNOWLEDGE**

The marine iguana (*Amblyrhynchus cristatus*), which spends long periods under water feeding on seaweed, relies on both salt glands and kidneys for homeostasis of its internal fluids. Describe how these organs together meet the particular osmoregulatory challenges of this animal's environment.

For selected answers, see Appendix A.